

S.No. : 201

BCS 3604

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 33228

Roll
No.

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B. Tech. Examination 2021-22

(Even Semester)

COMPILER DESIGN

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) What do you understand by Pass?
- (b) How many types of bottom up parsers?
- (c) What are advantages of SDT?
- (d) How many types of intermediate code generation?
- (e) What information is represented by symbol tables?

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- (f) How many types of loops in flow graph?
- (g) Discuss about DAG representation.
- (h) Discuss the boot strapping of cross compiler.

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Construct NFA equivalent $r = a * b$.
- (b) Consider the grammar :

$$S \rightarrow (L) \mid a$$

$$L \rightarrow L, S \mid S$$

Find Parse trees for the following sentences :

- (i) $(a, (a, a))$
- (ii) $(a, ((a, a), (a, a)))$
- (c) Check whether the grammar is LL(1)? Remove left recursion and then again verify whether it is LL(1)?

$$S \rightarrow A a \mid bG$$

$$L \rightarrow Ac \mid Sd \mid \epsilon$$

- (d) Design parsing table of LL(1) for the following grammar :

$$A \rightarrow A c B \mid c C \mid C$$

$$B \rightarrow b B \mid id$$

$$C \rightarrow C a B \mid B b B \mid B$$

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $5 \times 8 = 40$

3. (a) Consider the following grammar :

$$S \rightarrow (L) \mid a$$

$$L \rightarrow L, S \mid S$$

Parse the input string (a, (a, a)) using SR parser.

- (b) Construct parse tree, syntax tree and annotated parse tree for the input string is $5 * 6 + 7$.
- (c) Differentiate between synthesized translation and inherited translation.
4. (a) Write the quadruples, triples and indirect triples for the expression :
- $$-(a * b) + (c + d) - (a + b + c + d)$$
- (b) What is post fix translation? Explain it with a suitable example.

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- (c) Explain the following. Give example also :
- (i) Lexical phase errors
 - (ii) Syntactic phase errors
5. (a) Generate the code for following expression :
- $$(a - b) + c * (d/e)$$
- (b) Explain the following in the organization of the code optimizer :
- (i) Control flow analysis
 - (ii) Data flow analysis
 - (iii) Transformation
- (c) Explain the optimization of basic blocks. Also explain the DAG representation of basic blocks.
6. (a) Consider the following program code for program for computing dot product of two vector a and b of length 10 and partition it into basic block :
- ```
prod = 0;
i = 1;
do
{
 prod = prod + a[i] * b[i];
 i = i + 1;
} while (i <= 10)
```

- (b) Write the quadruples, triples and indirect for the following expression :

$$(x + y) * (y + z) + (x + y + z)$$

- (c) Write short notes on the following :
- (i) Context free grammars. Give examples of CFG.
  - (ii) Parse tree. Give an example of parse tree.

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